Vidyamandir Classes
Gurukul for IITJEE & Medical Preparation

illuminati - 2019

Advanced Mathematics Assignment-1B

ALGEBRA

Section 1	Single Correct Answer Type

Each of the following Question has 4 choices A, B, C & D, out of which ONLY ONE Choice is Correct.

1. Let $S_n = \frac{1}{4} + \frac{1.3}{4.6} + \frac{1.3.5}{4.6.8} + ...$ then $\lim_{n\to\infty} S_n =$

(A) $\frac{1}{2}$ **(B)** 1 **(C)** $\frac{3}{4}$ **(D)** 2

2. If $a_1 = 1$ & for $n > 1$, $a_n = a_{n-1} + \frac{1}{a_{n-1}}$, then a_{75} lies between :

(A) $(12, 15)$ **(B)** $(15, 18)$ **(C)** $(11, 12)$ **(D)** None of these

3. Let a series be defined such that its r^{th} term t_r is given by $t_r = (r-1)S_{r-1}$ for $r > 1$, where S_r is sum of its first r terms, also $t_1 > 0$. Then for $n > 1$, $(n \in N)$.

(A) $S_n = (n+1)!t_1$ **(B)** $(n+1)t_{n+1} + S_n = nS_{n+1}$

(C) $nt_{n+1} + S_n = (n+1)S_{n+1}$ **(D)** $(n+2)t_{n+1} + S_n = (n+1)S_{n+1}$

4. Let $1^P + 2^P + 3^P + ... + n^P = a_0 n^{P+1} + a_1 n^P + a_2 n^{P-1} .... + a_P n + a_{P+1}$, where $n, P \in N$. If $a_0 = \frac{1}{8}$, then $P =$

(A) 7 **(B)** 8 **(C)** 6 **(D)** None of these

5. If $f(x) = \left|a^2x^2 + b|x| + c\right|$ is non-differentiable at two points then the sum of the roots of the equation $a^2x^2 + bx + c = 0$ is always :

(A) positive **(B)** negative **(C)** zero **(D)** can't say

6. The complete set of non-zero values of k such that the equation $|x^2 - 10x + 9| = kx$ is satisfied by atleast one and atmost three values of x, is :

(A) $[4, \infty) \cup (-\infty, -16]$ **(B)** $[16, \infty) \cup (-\infty, -16]$

(C) $[4, \infty) \cup (-\infty, -4]$ **(D)** $[-16, 4]$

7. If the line $ax + by + c = 0$ lies in exactly two quadrants then of the set of quadratic equations, $x^2 + ax - ab = 0;\ x^2 + bx - bc = 0;\ x^2 + cx - ac = 0;$

(A) At least one has identical roots **(B)** At most two have identical roots

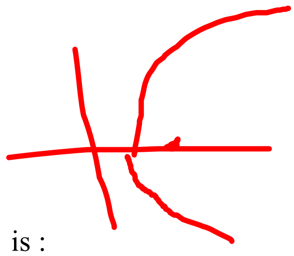
(C) Both (A) and (B) **(D)** Exactly one has identical roots

8. If z is a complex number such that $-\frac{\pi}{2} < \arg z \le \frac{\pi}{2}$, then which of the following inequality is true :

(A) $|z-\bar{z}| \le |z|(\arg z - \arg \bar{z})$ **(B)** $|z-\bar{z}| \ge |z|(\arg z - \arg \bar{z})$

(C) $|z-\bar{z}| < |z|(\arg z - \arg \bar{z})$ **(D)** None of these

9. If z be a complex number and $a_i, b_i, (i = 1, 2, 3)$ are real numbers, then the value of the determinant $\begin{vmatrix} a_1 z + b_1 \bar{z} & a_2 z + b_2 \bar{z} & a_3 z + b_3 \bar{z} \\ b_1 z + a_1 \bar{z} & b_2 z + a_2 \bar{z} & b_3 z + a_3 \bar{z} \\ b_1 z + a_1 & b_2 z + a_2 & b_3 z + a_3 \end{vmatrix}$ is equal to :



(A) $(a_1 a_2 a_3 + b_1 b_2 b_3)|z|^2$ **(B)** $|z|^2$

(C) 0 **(D)** None of these

10. The greatest positive argument of complex number satisfying $|z-4| = \operatorname{Re}(z)$ is :

(A) $\frac{\pi}{3}$ **(B)** $\frac{2\pi}{3}$ **(C)** $\frac{\pi}{2}$ **(D)** $\frac{\pi}{4}$

11. The complex number associated with the vertices A, B, C of the ΔABC are $e^{i\theta}, \omega, \bar{\omega}$ respectively (where $\omega, \bar{\omega}$ are the complex cube roots of unity and $\cos\theta > \operatorname{Re}(\omega)$), then the complex number of the point where angle bisector of A meets the circumcircle of the triangle, is :

(A) $-e^{i\theta}$ **(B)** $-e^{-i\theta}$ **(C)** $\omega\bar{\omega}$ **(D)** $\omega + \bar{\omega}$

12. If α is the root of equation $z^n + 2z^{n-1} + 3z^{n-2} + 12 - 18z = 0$, which lie inside $|z| = 1$, then :

(A) $|\alpha| > \frac{2}{3}$ **(B)** $|\alpha| < \frac{2}{3}$ **(C)** $|\alpha| = \frac{2}{3}$ **(D)** None of these

13. If z_1 and z_2 satisfy $|z-(3+2i)| = \operatorname{Re} z$, and $\arg\left(\frac{z_1 - 4i}{z_2 - 4i}\right) = \frac{\pi}{2}$ then $\arg\left(\frac{z_1 - (3+2i)}{z_2 - (3+2i)}\right)$ is equal to :

(A) 0 **(B)** π **(C)** $\pm\frac{\pi}{2}$ **(D)** None of these

14. If z is a complex number lying in the fourth quadrant of argand plane and $\left|\frac{kz}{k+1} + 2i\right| > \sqrt{2}$ for all real value of $k\,(k \ne -1)$, then range of $\arg(z)$ is :

(A) $\left(-\frac{\pi}{8}, 0\right)$ **(B)** $\left(-\frac{\pi}{6}, 0\right)$ **(C)** $\left(-\frac{\pi}{4}, 0\right)$ **(D)** None of these

15. The sum of the infinite series $\frac{1.3}{2} + \frac{3.5}{2^2} + \frac{5.7}{2^3} + \frac{7.9}{2^4} + ...\infty$ is :

(A) 32 **(B)** 23 **(C)** 29 **(D)** 30

16. The number of rational numbers (upto 9 digit after decimal) lying in the interval (2014, 2015) all whose digits after the decimal points are non-zero and are in non increasing order, is :

(A) $\sum_{i=1}^{9} {}^9P_i$ **(B)** $2^9 - 1$ **(C)** ${}^{18}C_9$ **(D)** $2^{10} - 1$

17. If $\sin^{-1}\left(\sin\left(\frac{2x^2+4}{1+x^2}\right)\right) < \pi - 3$, then x belongs to :

(A) $|x| > 1$ **(B)** $|x| \ge 1$ **(C)** $|x| < 1$ **(D)** none of these

18. If $\sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6}$ and $S_i = \sum_{k=1}^{\infty} \frac{i}{(36k^2-1)^i}$, then $S_1 + S_2$ is equal to :

(A) $\frac{\pi^2}{9} - \frac{1}{2}$ **(B)** $\frac{\pi^2}{12} + \frac{1}{2}$ **(C)** $\frac{\pi^2}{15} + \frac{1}{2}$ **(D)** $\frac{\pi^2}{18} - \frac{1}{2}$

19. The equation $2^{\frac{2\pi}{\cos^{-1}x}} - \left(a + \frac{1}{2}\right)2^{\frac{\pi}{\cos^{-1}x}} - a^2 = 0$ has only one real root then :

(A) $1 \le a \le 3$ **(B)** $a \ge 1, a \le -3$ **(C)** $1 < a < 3$ **(D)** None of these

20. If the roots of the equation $ax^2 + bx + c = 0$ are α, β then the roots of the equation $b^2cx^2 - ab^2x + a^3 = 0$ are :

(A) $\frac{1}{\alpha^2 + \alpha\beta}, \frac{1}{\beta^2 + \alpha\beta}$ **(B)** $\frac{1}{\alpha^3 + \alpha\beta}, \frac{1}{\beta^3 + \alpha\beta}$

(C) $\frac{1}{\beta^4 + \alpha\beta}, \frac{1}{\alpha^4 + \alpha\beta}$ **(D)** None of these

21. If α, β are the roots of the equation $x^2 - x + q = 0$ and $S_r = \alpha^r + \beta^r$, (where $r \in N \cup \{0\}$) then S_{k+1} is equal to :

(A) $1 - q(S_{k-1} + S_{k-2} + ... + S_0)$ **(B)** $1 + q(S_{k-1} + S_{k-2} + ... + S_0)$

(C) $(S_{k-1} + S_{k-2} + ... + S_0)$ **(D)** None of these

22. If $k + |k + z^2| = |z|^2 \ (k \in R^-)$ then argument of z is :

(A) 0 **(B)** π **(C)** $-\pi/2$ **(D)** none of these

23. $(x+m)^m - {}^mC_1(x+m-1)^m + {}^mC_2(x+m-2)^m - .... + (-1)^m . x^m$ is equal to :

(A) $(m+1)!$ **(B)** ${}^{m+1}P_m$ **(C)** $m!$ **(D)** ${}^mP_{m-2}$

24. In how many ways we can distribute 999 identical balls in 3 identical boxes ?

(A) 83667 **(B)** 83668 **(C)** 500500 **(D)** None of these

25. $8n$ players $P_1, P_2, P_3......., P_{8n}$ play a knock out tournament. It is known that all the players are of equal strength. The tournament is held in 3 rounds where the players are paired at random in each round. If it is given that P_1 wins in the third round. The probability that P_2 looses in the second round is :

(A) $\frac{n}{8n-1}$ **(B)** $\frac{n}{8n+1}$ **(C)** $\frac{2n}{4n-1}$ **(D)** None of these

26. Three persons A, B, C invite $2n$ guests to a party. All the people sit around a round table such that A, B, C have their fixed seats (there is at least one seat between any two fixed seats) and two particular guests does not sit together. Find the number of ways in which they can be seated ?

(A) $(4n^2 - 6n + 6)(2n-2)!$ **(B)** $(2n-3)2\cdot(2n-2)!$

(C) $(2n-2)!$ **(D)** None of these

27. Find the non negative integral solution of $kx + y + z + w = kr$, where k and $r \in I^+$.

(A) $\frac{(r+1)}{12}\left(2k^2r^2 + k^2r + 9kr + 12\right)$ **(B)** $\frac{(r+1)}{6}\left(2k^2r^2 + k^2r + 9kr + 12\right)$

(C) $\left(2k^2r^2 + k^2r + 9kr + 12\right)$ **(D)** None of these

28. Find the number of $2n$ digit numbers whose all digits are prime and sum of any two consecutive digits is also prime.

(A) 2^n **(B)** 2^{n+1} **(C)** 4^{n-1} **(D)** 2^{2n}

29. Evaluate $\sum_{k=0}^{n+1}\sum_{j=1}^{k+1}\sum_{i=1}^{k} {}^iC_{j-1}$ where $n > 2$.

(A) $2^{n+3} - 2n - 6$ **(B)** $2^{n+3} - 2n - 4$ **(C)** $2^{n+2} - 4n - 2$ **(D)** None of these

30. for $m \geq 1$, $\frac{{}^mC_n}{2} + \frac{{}^{m+1}C_n}{2^2} + \frac{{}^{m+2}C_n}{2^3} + \text{.........}$ to $\infty =$

(A) $\sum_{r=0}^{n} {}^mC_r$ **(B)** $\sum_{r=1}^{n} {}^mC_r$ **(C)** $\sum_{r=m}^{n} {}^rC_m$ **(D)** None of these

Section 2	**Link Comprehension Type**

Each of the following Question has 4 choices A, B, C & D, out of which ONLY ONE Choice is Correct.

Paragraph for Questions 31 - 33

If $\left(1 + px + x^2\right)^n = 1 + a_1x + a_2x^2 + ... + a_{2n}x^{2n}$, then

31. Which of the following is true for $1 < r < 2n$?

(A) $(np + pr)a_r = (r+1)a_{r+1} + (r-1)a_{r-1}$

(B) $(np - pr)a_r = (r+1)a_{r+1} + (r-1-2n)a_{r-1}$

(C) $(np - pr)a_r = (r+1)a_{r+1} + (r-1-n)a_{r-1}$

(D) $(2np + pr)a_r = (r+1+n)a_{r+1} + (r+1-n)a_{r-1}$

32. The remainder obtained, when $a_1 + 5a_2 + 9a_3 + 7a_4 + \text{........} + (8n-3)a_{2n}$, is divided by $(p+2)$ is :

(A) 1 **(B)** 2 **(C)** 3 **(D)** 0

33. The value of $a_1 + 3a_2 + 5a_3 + 7a_4 + \text{......} + (4n-1)a_{2n}$, when $p = -3$ and $n \in$ even, is :

(A) n **(B)** $2n-1$ **(C)** $2n-2$ **(D)** $2n$

Paragraph for Questions 34 - 36

A triangle is called an integral triangle if all its sides are integers. If a, b, and c are sides of an integral triangle, then we an assume that $a \leq b \leq c$, because any other permutation of a, b, and c will yield the same result. Now, answer the following question.

34. The number of integral isosceles or equilateral triangle's, none of whose side exceeds 4, is :

(A) 9 **(B)** 10 **(C)** 11 **(D)** 12

35. The number of integral isosceles or equilateral triangles, none of whose side exceeds $2p$, is :

(A) $4p^2-2p$ **(B)** $2p^2+2p$ **(C)** $3p^2$ **(D)** $\frac{3p^2}{2}+3p$

36. If c is fixed and odd, then number of integral isosceles or equilateral triangle's whose sides are $a,b,c\left(a\le b\le c\right)$ must be :

(A) $\frac{2c-1}{2}$ **(B)** $\frac{2c+1}{2}$ **(C)** $\frac{3c+1}{2}$ **(D)** $\frac{3c-1}{2}$

Paragraph for Questions 37 - 39

Inside a triangle ABC let there exist a unique point P such that $\angle PAB = \angle PBC = \angle PCA = \alpha$. Let $PA = x$, $PB = y$, $PC = z$, Δ = area of triangle ABC, a, b, c are the sides opposite the angle A, B, C respectively.

37. $\tan\alpha$ equals :

(A) $\frac{\Delta}{a^2+b^2+c^2}$ **(B)** $\frac{2\Delta}{a^2+b^2+c^2}$ **(C)** $\frac{4\Delta}{a^2+b^2+c^2}$ **(D)** $\frac{1}{\Delta\left(a^2+b^2+c^2\right)}$

38. $\text{cosec}^2\alpha$ equals :

(A) $\sin^2A+\sin^2B+\sin^2C$ **(B)** $\text{cosec}^2A+\text{cosec}^2B+\text{cosec}^2C$

(C) $\cos^2A+\cos^2B+\cos^2C$ **(D)** $\sec^2A+\sec^2B+\sec^2C$

39. Which is always true :

(A) $0<\alpha\le 30°$ **(B)** $30°<\alpha\le 45°$

(C) $45°<\alpha\le 90°$ **(D)** $90°<\alpha<180°$

Section 3	**Multiple Correct Answer Type**

Each of the following Question has 4 choices A, B, C & D, out of which ONE OR MORE Choices may be Correct.

40. Number of triangles which can be formed by joining vertices of a regular polygon of $n(>5)$ sides such that no side is common with the side of polygon is equal to :

(A) $\frac{n}{n-3}\,{}^{n-3}C_3$ **(B)** ${}^nC_3-n-n\left(n-4\right)$

(C) ${}^{n-4}C_2+{}^{n-3}C_3$ **(D)** ${}^{n+2}C_3$

41. Assume a group of r has to be selected from n students (r can vary from 1 to n). From a selected group a treasurer, a secretary and a chairperson have to be elected (may be same person or may be different). The number of ways in which this can be done will be :

(A) $\sum_{r=1}^{n}r^3\,{}^nC_r$ **(B)** $n2^{n-1}+3n\left(n-1\right)2^{n-2}+n\left(n-1\right)\left(n-2\right)2^{n-3}$

(C) $\frac{\sum_{r=1}^{n}r^3\,{}^nC_r}{3}$ **(D)** $\sum_{r=1}^{n}r\cdot\left(r-1\right)\cdot\left(r-2\right){}^nC_r$

42. If $S_n=\frac{\left(1^2+1-1\right)\cdot 1!}{1^2+3\cdot 1+2}+\frac{\left(2^2+2-1\right)\cdot 2!}{2^2+3\cdot 2+2}+.....+\frac{\left(n^2+n-1\right)\cdot n!}{\left(n^2+3n+2\right)}$. Then :

(A) $S_5=\frac{179}{2}$ **(B)** $S_7=\frac{9119}{2}$ **(C)** $S_5=\frac{181}{2}$ **(D)** $S_7=\frac{9117}{2}$

43. If a & b are positive numbers, then the equation $\frac{1}{x}+\frac{1}{x-a}+\frac{1}{x+b}=0$ has :

(A) Two real roots **(B)** one root lying between $\left(\frac{a}{3},\frac{2a}{3}\right)$

(C) two roots lying between $\left(\frac{a}{3},\frac{2a}{3}\right)$ **(D)** one root lying between $\left(\frac{-2b}{3},\frac{-b}{3}\right)$

44. If ${}^{100}C_6+4\cdot{}^{100}C_7+6\,{}^{100}C_8+4\cdot{}^{100}C_9+{}^{100}C_{10}$ has the value equal to ${}^{x}C_y$, then the value of $(x+y)$ can be :

(A) 114 **(B)** 115 **(C)** 198 **(D)** 199

45. The value of the constant term in the binomial $\left(x^2+\frac{1}{x^2}-2\right)^{10}$ is also equal to :

(A) number of different dissimilar terms in $(x_1+x_2+.......+x_{10})^{10}$

(B) $\left({}^{10}C_0\right)^2+\left({}^{10}C_1\right)^2+...........+\left({}^{10}C_{10}\right)^2$

(C) Coefficient of x^{10} in $(1-x)^{20}$

(D) number of linear arrangements of 20 things of which 10 alike to one kind and rest all are different, taken all at a times.

46. The number of ways in which five different books to be distributed among 3 persons, so that each person gets at least one book, is equal to the number of ways in which ?

(A) 5 persons are allotted 3 different residential flats, so that each person is allotted at most one flat and no two persons are allotted the same flat.

(B) Number of parallelograms formed by one set of 6 parallel lines and other set of 5 parallel liens that goes in other direction.

(C) 5 different toys are to be distributed among 3 children , so that each child get atleast one toy.

(D) 3 mathematics professors are assigned five different lectures to be delivered , so that each professor gets atleast one lecture.

47. Considering the principal values of inverse trigonometric functions $2\tan^{-1}(-3)$ is equal to :

(A) $-\pi-\cos^{-1}\left(\frac{4}{5}\right)$ **(B)** $-\pi+\cos^{-1}\left(\frac{4}{5}\right)$

(C) $-\frac{\pi}{2}-\tan^{-1}\left(\frac{4}{3}\right)$ **(D)** $-\frac{\pi}{2}-\sin^{-1}\left(\frac{4}{5}\right)$

48. Let x_1, x_2, x_3, x_4 be four non-zero numbers satisfying the equation $\tan^{-1}\frac{a}{x}+\tan^{-1}\frac{b}{x}+\tan^{-1}\frac{c}{x}+\tan^{-1}\frac{d}{x}=\frac{\pi}{2}$. Then :

(A) $x_1+x_2+x_3+x_4=a+b+c+d$

(B) $\frac{1}{x_1}+\frac{1}{x_2}+\frac{1}{x_3}+\frac{1}{x_4}=0$

(C) $x_1\,x_2\,x_3\,x_4=abcd$

(D) $(x_2+x_3+x_4)(x_3+x_4+x_1)(x_4+x_1+x_2)(x_1+x_2+x_3)=abcd$

49. If $\sin^{-1}(a^2x^2+b^2y^2)+\cos^{-1}|ax+by|=\pi$ where $ab<0$, then :

(A) $ab=-\frac{1}{2},\ xy=1$ **(B)** $ax=\pm\frac{1}{\sqrt{2}},\ by=\pm\frac{1}{\sqrt{2}}$

(C) $ay=-\frac{1}{2},\ bx=1$ **(D)** $a=b=x=y=1$

50. Let a sequence $t_1, t_2, t_3, ...$ be defined such that $t_k=t_{k-1}+t_{k+1}$ for $k\geq 2$. If $t_1+t_{10}=-10$ and $\sum_{i=1}^{10} t_i=20$, then :

(A) $t_2+t_9=20$ **(B)** $\sum_{i=3}^{8} t_i=10$ **(C)** $t_4-t_7=5$ **(D)** $t_4+t_7=5$

51. If p and q are odd integers, then the equation $x^2+2px+2q=0$.

(A) has no integral roots **(B)** has no rational roots

(C) has no irrational roots **(D)** has no imaginary roots.

52. If the equation $ax^2+bx+c=0 (a<0)$ has two roots α and β such that $\alpha<-3$ and $\beta>3$, then :

(A) $9a+3|b|+c>0$ **(B)** $c>0$

(C) $4a+2|b|+c>0$ **(D)** None of these

53. If the equation $\sin^2 x-a\sin x+b=0$ has only one solution in $(0,\pi)$ then which of the following statements are correct ?

(A) $a\in(-\infty, 1]\cup[2, \infty)$ **(B)** $b\in(-\infty, 0]\cup[1, \infty)$

(C) $a=1+b$ **(D)** $b=1+a$

54. If sum of the arguments of all the roots of equation $x^n=k$ is π, then it can be true when :

(A) n is even and $k>0$ **(B)** n is even and $k<0$

(C) n is odd and $k>0$ **(D)** n is odd and $k<0$

Section 4	**Match Matrix Type**

I.

	Column-I		**Column - II**		**Column -III**
(I)	If $A=\sum_{r=1}^{n} r^2,\ B=\sum_{m=1}^{n}\sum_{r=1}^{m} r-\frac{1}{2}\sum_{r=1}^{n} r$, then $\frac{A}{B}$ will take	**(i)**	Minimum value	**(P)**	6
(II)	For positive a, b, c: $\frac{a(b^2+c^2)+b(c^2+a^2)+c(a^2+b^2)}{abc}$ will take	**(ii)**	Maximum value	**(Q)**	1
(III)	If $x+y+z=1$, then $\frac{2x^2}{y+z}+\frac{2y^2}{z+x}+\frac{2z^2}{x+y}$ will take	**(iii)**	Value equal to	**(R)**	2
(IV)	If $\frac{1}{xy}+\frac{1}{yz}+\frac{1}{zx}=3$, where $x, y, z\in N$, then $x+y+z$	**(iv)**	Can take value	**(S)**	3

55. Which one option is the only CORRECT combination ?

(A) (I) (i) (R) **(B)** (I) (iv) (R) **(C)** (II) (iv) (S) **(D)** (III) (i) (R)

56. Which one option is the only CORRECT combination ?

(A) (IV) (iii) (S) **(B)** (III) (ii) (R) **(C)** (IV) (ii) (S) **(D)** (II) (ii) (R)

57. Which one option is the only INCORRECT combination ?

(A) (II) (ii) (P) **(B)** (II) (iv) (P) **(C)** (III) (iv) (Q) **(D)** (IV) (iv) (S)

II. Let $f(x+y) = f(x)f(y)\ \forall\ x, y \in R$ and $f(1) = 2$. If in triangle ABC, $a = f(3)$, $b = f(1) + f(3)$ and $c = f(2) + f(3)$; r, R denote the in radius and circum radius of triangle ABC, then :

Column-I		**Column - II**		**Column -III**	
(I)	$\frac{R}{r}$	**(i)**	$\geq$	**(P)**	$\frac{16}{9}$
(II)	$\tan^2 A$	**(ii)**	$>$	**(Q)**	$\frac{7}{9}$
(III)	$\sec^2 A$	**(iii)**	$\leq$	**(R)**	$\frac{16}{7}$
(IV)	r	**(iv)**	$<$	**(S)**	$\sqrt{7}$

58. Which one option is the only CORRECT combination ?

(A) (I) (i) (S) **(B)** (I) (iv) (R) **(C)** (II) (iv) (Q) **(D)** (III) (iii) (P)

59. Which one option is the only INCORRECT combination ?

(A) (IV) (iv) (S) **(B)** (IV) (ii) (P) **(C)** (III) (iv) (R) **(D)** (II) (i) (Q)

60. Which one option is the only CORRECT combination ?

(A) (III) (iii) (Q) **(B)** (I) (i) (R) **(C)** (I) (iv) (Q) **(D)** (IV) (iv) (S)

Section 5	**Single Integer Answer Type**

Each of the following question has an integer answer between 0 and 9.

61. The sum of the infinite series $\frac{3}{1.2.4} + \frac{4}{2.3.5} + \frac{5}{3.4.6} + \ldots = \frac{p}{q}\,(GCF(p,q)=1)$ (p, q) = 1) then $|p-q| =$

62. Let $\sqrt{1 + \frac{1}{1^2} + \frac{1}{2^2}} + \sqrt{1 + \frac{1}{2^2} + \frac{1}{3^2}} + \ldots\ldots + \sqrt{1 + \frac{1}{1999^2} + \frac{1}{2000^2}} = p + \frac{p}{q}$.

$\left(0 < \frac{p}{q} < 1 \text{ and } GCF(p,q) = 1\right)$ Then $|p-q| =$

63. If $az^2 + bz + 1 = 0$, where $a, b \in c$ and $|a| = \frac{1}{2}$, have a root α such that $|\alpha| = 1$ and $|a\bar{b} - b| = \frac{p}{q}$ $(GCF\,(p, q) = 1)$ then $|p-q| =$

1

64. The complex number z satisfying $|z+2+1| + |z-2+i| = 4$, $0 \leq \arg(z+2+2i) \leq \pi/4$ and $\frac{3\pi}{4} \leq \arg(z-2+2i) \leq \pi$ will lie on a line segment of length.

65. The value of $\frac{n}{n-3} \, {}^{n-3}C_3$ then x is :

66. $\lim\limits_{x \to 0} \left(\sum\limits_{r=0}^{n} (-1)^r \cdot {}^nC_r \cdot \left(\sum\limits_{k=0}^{n-r} {}^{n-r}C_k \cdot x^k \cdot 2^k \right) \cdot \left(x^2 - x\right)^r \right)^{\frac{1}{x}} = e^{n\lambda}$, then λ is equal to :

67. If $S = \sum\limits_{k=1}^{\infty} \frac{1}{k^2}$ and $t = \sum\limits_{k=1}^{\infty} \frac{(-1)^{k+1}}{k^2}$, then the value of $\frac{s}{t}$ is equal to :

68. The remainder when the number $555^{555} + 666^{666} + 888^{888} + 999^{999}$ is divided by 7 is :

69. All the five digit numbers in which each successive digit exceeds it predecessor are arranged in the increasing order of their order of their magnitudes. The 97th number is n, whose some of digits is k, then $\frac{k}{3}$ is :

70. If the angle at the vertex of an isosceles triangle having the maximum area for the given length l of the median to one of its equal sides, is x then 5cosx is equal to________.

71. Let $S_n = \left(\frac{2}{1 \times 2}\right) + \left(\frac{5}{2 \times 3} \times 2\right) + \left(\frac{10}{3 \times 4} \times 2^2\right) + \ldots\ldots$..upto n terms. Then, the number of values of n (where $n \in [2, 15)$) for which S_n is an integer.

72. Let $a_n = 16, 4, 1, \ldots\ldots$ be a geometric sequence. Define b_n as product of first n terms. Then value of $\frac{1}{8} \sum\limits_{n=1}^{\infty} \sqrt[n]{b_n}$ is ________. 4

73. If the equation $x^2 + nx + n = 0, n \in I$, has integral roots, then the number integral values $n^2 - 4n$ can assume : 4

74. Least natural number $'a'$ for which $x + ax^{-2} > 2 \;\forall\; x \in (0, \infty)$ is : 2

75. The roots of $x^3 + px^2 + qx - 9 = 0$ are each one more than the roots of $x^3 - ax^2 + bx - c = 0$, where $a, b, c, p, q \in R$. Then $a + b + c$ is equal to : 8

ANSWERS KEY

Advanced Mathematics Assignment-1B | 2019 | Algebra

Section 1	**Single Correct Answers Type**

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
B	A	D	A	C	A	D	D	C	D	D	A	B	C	B
16	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
C	C	D	B	A	A	C	C	A	D	A	A	B	A	A

Section 2	**Link Comprehension Type**

31	32	33	34	35	36	37	38	39
B	C	D	D	C	D	C	B	A

Section 3	**Multiple Correct Answer Type**

40	41	42	43	44	45	46	47	48	49	50
ABC	AB	AB	ABD	AC	BC	BCD	BCD	BCD	ABC	ABD

51	52	53	54
AB	ABC	ABC	AD

Section 4	**Matrix Match Type**

55	56	57	58	59	60
B	A	A	D	A	B

Section 5	**Single Integer Answer Type**

61	62	63	64	65	66	67	68	69	70
7	1	1	2	2	3	2	2	9	4

71	72	73	74	75
6	4	1	2	9